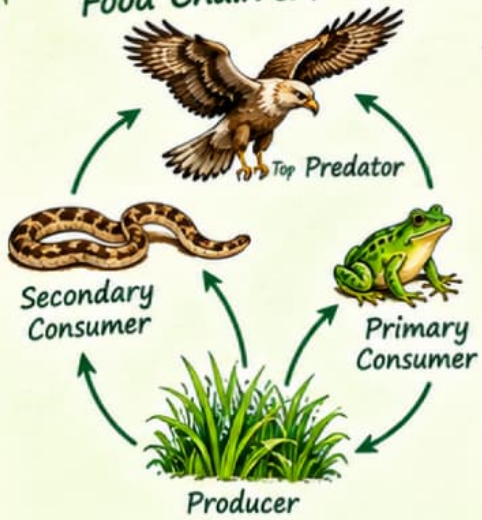
ABN SEAL COLLEGE
COOCHBEHAR
PANCHANAN BARMA UNIVERSITY
COOCHBEHAR

Ecosystem

B.Sc. 2nd Semester | CBPBU

Created by
Bikram Singha



Components of Ecosystem

Biotic
- Producers
- Consumers
- Decomposers

Abiotic
- Sunlight
- Water
- Soil
- Air





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Ecosystem → Functional unit of nature where living organism interact ⊂ non living elements.

- Term given by → Arthur George Tansley → In 1935 → British ecologist
- Eco + System → Eco → Habitat; System → complex set of inter-connected components
- Devided into two types → Terrestial ecosystem; aquatic ecosystem.

∅ Inputs & Outputs of an ecosystem

Ecosystem is not isolated, they continuously exchange both living & non living matter ⊂ their sorrounding ecosystem.

Inputs: Sorrounding environment → to → ecosystem

Outputs: ecosystem → to → Sorrounding environment

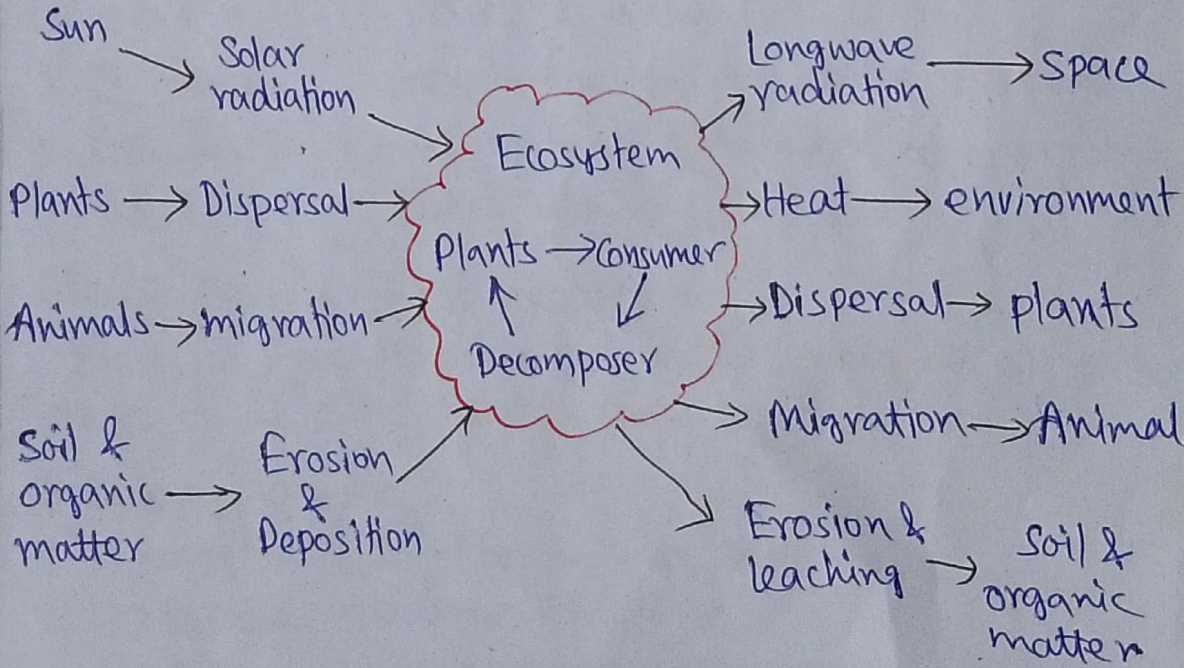
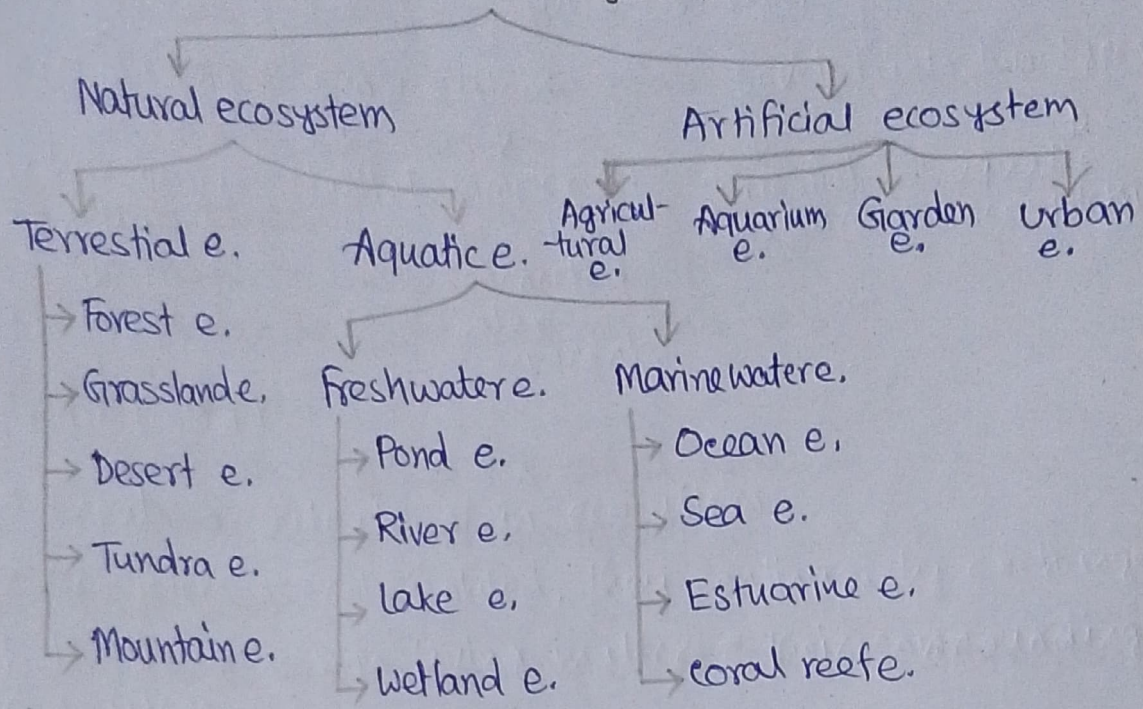


Fig: Schematic diagram of inputs & outputs of an ecosystem.

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Types of ecosystem



Stratification in ecosystem-

> Stratification means the arrangement of organisms into different layer/zones in an ~~ee~~ ecosystem according to environmental conditions such as light, temperature humidity & availability of nutrients. each layer supports different kind of plants & animals.

> It commonly seen in forest & aquatic ecosystems.

> Stratification is the vertical distribution of organisms into different layers within the ecosystem.

Stratification in Forest ecosystem

> forests show clean vertical ~~er~~ layering because plants of different heights grow together.

① Canopy layer-

> uppermost layer formed by tall trees

> Recieves maximum sunlight.

> Bird, monkeys & insects are common here.

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② Understory layer-
> located below the canopy
> Contains small trees & shrubs.
> recieves less sunlight.

③ Shrub layer-
> made of bushes & woody plants.
> Moderate light reaches this layer.

④ Herb layer-
> consists of grasses & small herbs.
> Near the ground surface.

⑤ Forest layer-
> lowest layer containing dead leaves, decomposers, fungi & microorganisms.
> Decomposition & nutrient recycling occur here.

Stratification in Aquatic ecosystem-
Water bodies also show stratification based on light penetration & depth.

① Littoral zone-
> Near the shore.
> Rich in plants & animals.

② ~~Liter~~ Limnetic zone-
> Open water region recieving sunlight.
> Phytoplankton & fishes are ~~abo~~ abundant.

③ Profundal zone-
> ~~Be~~ Deep water region with little light.
> Fewer organisms present.

④ Benthic zone-
> Botton region of the water body.
> Contains decomposer & bottom-dwelling organisms.

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